

ICIS-Air Data Structure

1 Overview

The EPA’s Integrated Compliance Information System for Air (ICIS-Air) provides 10 downloadable tables covering the universe of Clean Air Act regulated facilities. All tables share `PGM_SYS_ID` as the primary join key, which identifies a regulated facility. A secondary identifier, `REGISTRY_ID`, links to the Facility Registry Service (FRS) and identifies a physical site. Multiple `PGM_SYS_ID`s can map to one `REGISTRY_ID` when a single site has multiple separately regulated facilities.

2 Table Inventory

Table	Grain	Rows	Key Fields
FACILITIES	1 row per facility	279,211	Static characteristics
PROGRAMS	Facility $\times$ program	456,601	Program enrollment, dates
PROGRAM_SUBPARTS	Facility $\times$ subpart	190,570	MACT/NSPS subpart codes
POLLUTANTS	Facility $\times$ pollutant	976,479	Regulated pollutants
FCES_PCES	Evaluation event	1,802,044	Inspection type, date
VIOLATION_HISTORY	Violation event	101,147	HPV/FRV, dates, resolution
FORMAL_ACTIONS	Formal enforcement event	105,656	Action type, penalty
INFORMAL_ACTIONS	Informal enforcement event	336,410	NOV/warning, date
STACK_TESTS	Stack test event	646,332	Pass/fail, pollutants
TITLEV_CERTS	Certification event	2,563,435	Deviation flag

Table 1: ICIS-Air tables. All tables key on `PGM_SYS_ID`.

3 How the Tables Connect

Two types of tables

Static tables describe time-invariant (or slowly changing) characteristics of a facility:

- **FACILITIES**: master roster. One row per `PGM_SYS_ID`. Contains location, industry (NAICS/SIC), emissions classification (major/minor), and operating status.
- **PROGRAMS**: which CAA programs a facility is enrolled in (CAATVP = Title V, CAASIP = SIP, CAAMACT = MACT, etc.) with enrollment dates. Multiple rows per facility—one per program.
- **PROGRAM_SUBPARTS**: regulatory subpart detail within each program (e.g., which MACT subpart applies). Joins to PROGRAMS on `PGM_SYS_ID + PROGRAM_CODE`.
- **POLLUTANTS**: which pollutants a facility is regulated for. Multiple rows per facility—one per pollutant.

Event tables record time-stamped regulatory interactions:

- **FCES_PCES**: compliance evaluations (inspections). Each row is one evaluation event with an `ACTIVITY_ID`, type code (FCE on-site = `F00`, FCE off-site = `FF0`; 9 PCE codes), and date.
- **VIOLATION_HISTORY**: discovered violations. Each row is one violation with type (HPV or FRV), determination date, and resolution date (if resolved).
- **FORMAL_ACTIONS**: formal enforcement (administrative orders, judicial actions). Contains penalty amounts and settlement dates.
- **INFORMAL_ACTIONS**: informal enforcement (notices of violation = `NOV`, warning letters = `DAWL`). Contains achieved date.
- **STACK_TESTS**: emissions testing events with pass/fail status and pollutant codes.
- **TITLEV_CERTS**: annual Title V compliance certifications. Contains `FACILITY_RPT_DEVIATION_FLAG` (Y/N) indicating whether the facility self-reported deviations.

Cross-table linkages beyond `PGM_SYS_ID`

The event tables each assign their own `ACTIVITY_ID`, but these IDs are not shared across tables. A formal action's `ACTIVITY_ID` does not appear in the violation table, and vice versa. The tables do not record which enforcement action responded to which violation, or which evaluation discovered which violation.

The **Pipeline** dataset (`PIPELINE_CAA_00_COMPLETE.csv`) fills this gap by pre-joining violations to their linked evaluations and enforcement actions. It uses `VIOL_ACTIVITY_ID`, `EA_ACTIVITY_ID`, and `Eval_ACTIVITY_ID` to create the linkage. See the Pipeline Brief for details.

4 Key Relationships and Caveats

1. **PGM_SYS_ID vs. REGISTRY_ID**: `PGM_SYS_ID` identifies a regulated facility; `REGISTRY_ID` identifies a physical site. Multiple `PGM_SYS_ID`s can therefore share one `REGISTRY_ID` when a site contains separately regulated facilities.
2. **No direct violation-to-enforcement link**: the individual ICIS-Air tables do not record which enforcement action responded to which violation. A facility with 3 violations and 2 formal actions in the same year has no table-level information about which actions addressed which violations. The Pipeline dataset provides this linkage.
3. **Date field choice matters**: violations carry multiple date fields (`HPV_DAYZERO_DATE`, `EARLIEST_FRV_DETERM`, `SETTLEMENT_ENTERED_DATE`, etc.), so the choice of field affects which period an event is assigned to. Formal actions use `SETTLEMENT_ENTERED_DATE`, which may lag the initial filing by months or years.
4. **Evaluation type codes are abbreviations**: FCE on-site = `F00`, FCE off-site = `FF0`. PCE codes include `PCE`, `PFF`, `PFR`, `POC`, `POF`, `POI`, `POM`, `POR`, `POV`. These are not self-documenting; they are documented in EPA's ICIS-Air data dictionary.
5. **Event tables are not mutually exclusive**: a single regulatory interaction can generate records in multiple tables. An FCE that discovers a violation that leads to a formal action with a penalty produces rows in `FCES_PCES`, `VIOLATION_HISTORY`, and `FORMAL_ACTIONS`—so the same underlying interaction appears in several tables.
6. **Certifications are Title V only**: `TITLEV_CERTS` covers only Title V facilities. For non-Title V analyses, this table contributes nothing, and the self-reported deviation flag is unavailable.